

Отчёт по лабораторной работе 7

Адресация IPv4 и IPv6. Настройка DHCP

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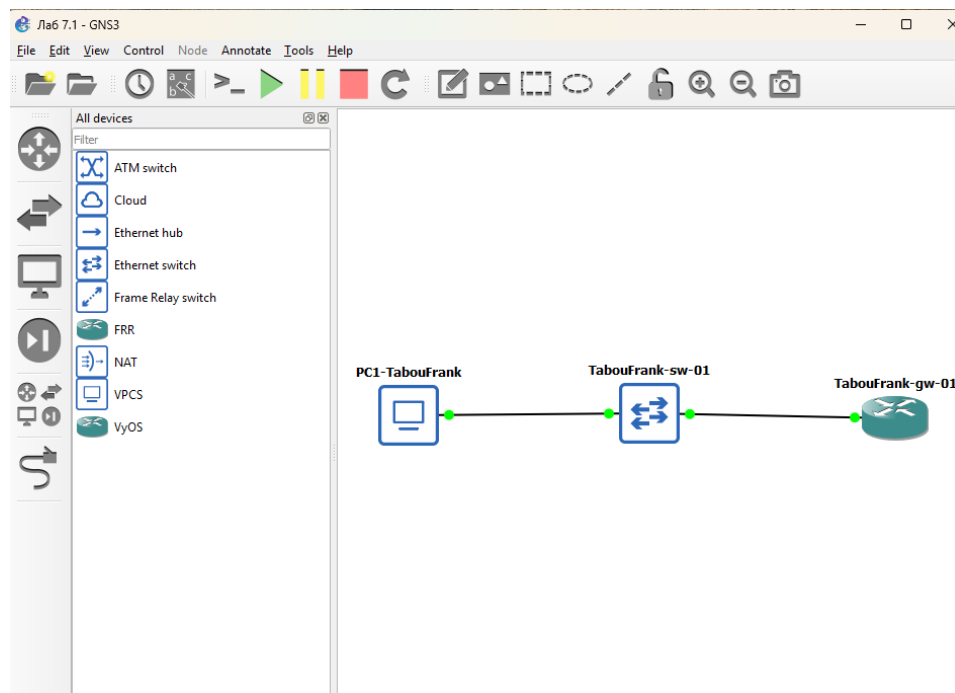


Рис. 1: Топология лабораторного стенда в GNS3

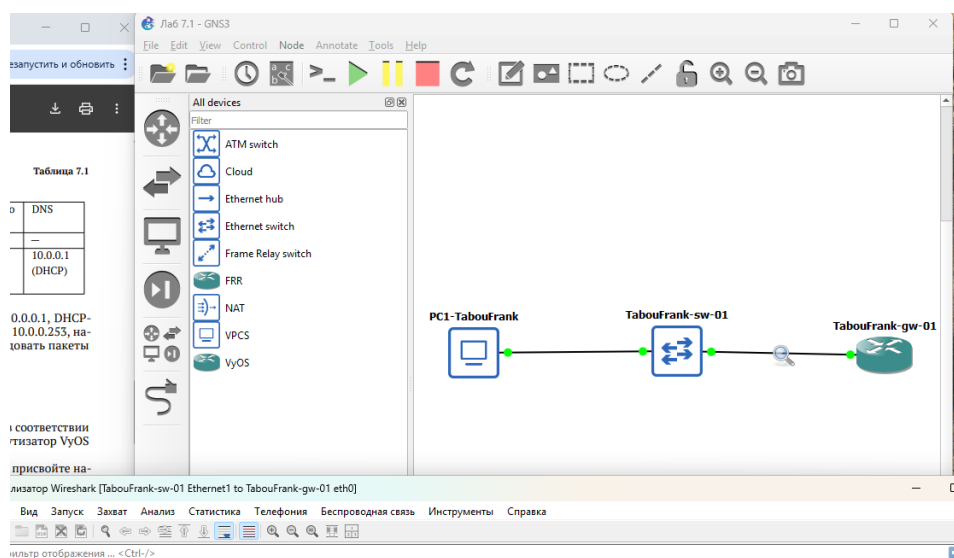


Рис. 2: Настройка IPv4-адреса интерфейса eth0

```

Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ install image
You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$ configure
[edit]
vyos@vyos# set ssystem host-name TabouFrank-gw-01

Configuration path: [ssystem] is not valid
Set failed

[edit]
vyos@vyos# set system host-name TabouFrank-gw-01
[edit]
vyos@vyos# set system domain-name TabouFrank.net
[edit]
vyos@vyos# set system login user TabouFrank authentication plaintext-password
TabouFrank

Configuration path: system login user TabouFrank [authentication] is not val
id
Set failed

[edit]
vyos@vyos# set system login user TabouFrank authentication plaintext-password d
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$

```

Рис. 3: Настройка DHCP-сервера на VyOS

```

Welcome to VyOS - TabouFrank-gw-01 ttyS0

TabouFrank-gw-01 login: TabouFrank
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
TabouFrank@TabouFrank-gw-01:~$

```

Рис. 4: Получение IP-адреса по DHCP на PC1

```

Welcome to VyOS - TabouFrank-gw-01 ttyS0

TabouFrank-gw-01 login: TabouFrank
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
TabouFrank@TabouFrank-gw-01:~$ configure
[edit]
TabouFrank@TabouFrank-gw-01# delete system login user vyos
[edit]
TabouFrank@TabouFrank-gw-01# commit
[edit]
TabouFrank@TabouFrank-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01# █

```

Рис. 5: Проверка IPv4-конфигурации и ping

```

Welcome to VyOS - TabouFrank-gw-01 ttyS0

TabouFrank-gw-01 login: TabouFrank
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
TabouFrank@TabouFrank-gw-01:~$ configure
[edit]
TabouFrank@TabouFrank-gw-01# delete system login user vyos
[edit]
TabouFrank@TabouFrank-gw-01# commit
[edit]
TabouFrank@TabouFrank-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24
[edit]
nk domain-name TabouFrank.net

Configuration path: service dhcp-server [shared-network-nae] is not valid
Set failed

[edit]
nk domain-name TabouFrank.net et service dhcp-server shared-network-nae TabouFrank
nk domain-name TabouFrank.net et service dhcp-server shared-network-name TabouFrank
[edit]
nk name-server 10.0.0.1-01# set service dhcp-server shared-network-name TabouFrank
[edit]
nk subnet 10.0.0.0/24 default-router 10.0.0.1 server shared-network-name TabouFrank
[edit]
nk subnet 10.0.0.0/24 range hosts start 10.0.0.2 server shared-network-name TabouFrank
[edit]
nk subnet 10.0.0.0/24 range hosts stop 10.0.0.253
[edit]
TabouFrank@TabouFrank-gw-01# █

```

Рис. 6: Статистика DHCP-сервера

```

[edit]
TabouFrank@TabouFrank-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24

Configuration path: [interfaces ethernet eth0 address 10.0.0.1/24] already exists

[edit]
TabouFrank@TabouFrank-gw-01# save
Warning: you have uncommitted changes that will not be saved.

Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01# commit

Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
[ service dhcp-server ]
None of the configured subnets have an appropriate primary IP address on
any broadcast interface configured, nor was there an explicit listen-
address configured for serving DHCP relay packets!

[[service dhcp-server]] failed
Commit failed
[edit]
TabouFrank@TabouFrank-gw-01# address dhcp

Invalid command: [address]

[edit]
TabouFrank@TabouFrank-gw-01# show configuration commands |match eth0

Configuration path: [configuration] is not valid
Show failed

[edit]
TabouFrank@TabouFrank-gw-01# show configuration commands | match eth0

Configuration path: [configuration] is not valid
Show failed

[edit]
TabouFrank@TabouFrank-gw-01# delete interfaces ethernet eth0 address dhcp
[edit]
TabouFrank@TabouFrank-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24

Configuration path: [interfaces ethernet eth0 address 10.0.0.1/24] already exists

[edit]
TabouFrank@TabouFrank-gw-01# show configuration commands | match eth0

Configuration path: [configuration] is not valid
Show failed

[edit]
TabouFrank@TabouFrank-gw-01# commit
[edit]
TabouFrank@TabouFrank-gw-01#

```

Рис. 7: Журнал работы DHCP-сервера

```

TabouFrank@TabouFrank-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01# exit
exit
TabouFrank@TabouFrank-gw-01:~$ show dhcp server statistics
Pool      Size    Leases  Available  Usage
-----
TabouFrank    252      0      252    0%
TabouFrank@TabouFrank-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start  Lease expiration  Remaining  Pool  Hostname
-----
TabouFrank@TabouFrank-gw-01:~$

```

Рис. 8: Анализа DHCP

```
TabouFrank@TabouFrank-gw-01:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L Description
-----
eth0           10.0.0.1/24      u/u
eth1           -                u/D
eth2           -                u/D
lo             127.0.0.1/8      u/u
              ::1/128
TabouFrank@TabouFrank-gw-01:~$
```

Рис. 9: Расширенная топология лабораторного стенда

```
VPCS> ip dhcp -d
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = TabouFrank.net

Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
```

Рис. 10: Настройка IPv6-адресов на интерфейсах VyOS

```
VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.0.0.2/24
GATEWAY    : 10.0.0.1
DNS        : 10.0.0.1
DHCP SERVER: 10.0.0.1
DHCP LEASE : 86264, 86400/43200/75600
DOMAIN NAME: TabouFrank.net
MAC        : 00:50:79:66:68:00
LPORT      : 10054
RHOST:PORT : 127.0.0.1:10055
MTU        : 1500

VPCS> ping 10.0.0.1 -c 2

84 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=3.137 ms
84 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=2.496 ms

VPCS>
```

Рис. 11: Настройка Router Advertisements и DHCPv6 Stateless

```
TabouFrank@TabouFrank-gw-01:~$ show dhcp server statistics
Pool      Size    Leases    Available    Usage
-----
TabouFrank 252      1          251          0%
TabouFrank@TabouFrank-gw-01:~$ show dhcp server statistics
Pool      Size    Leases    Available    Usage
-----
TabouFrank 252      1          251          0%
TabouFrank@TabouFrank-gw-01:~$ show dhcp server leases
IP address    Hardware address    State    Lease start    Lease expiration    Remaining    Pool    Hostname
-----
10.0.0.2      00:50:79:66:68:00    active   2026/01/03 23:50:18    2026/01/04 23:50:18    23:51:55    TabouFrank    VPCS
TabouFrank@TabouFrank-gw-01:~$
```

Рис. 12: Просмотр конфигурации DHCPv4 и DHCPv6

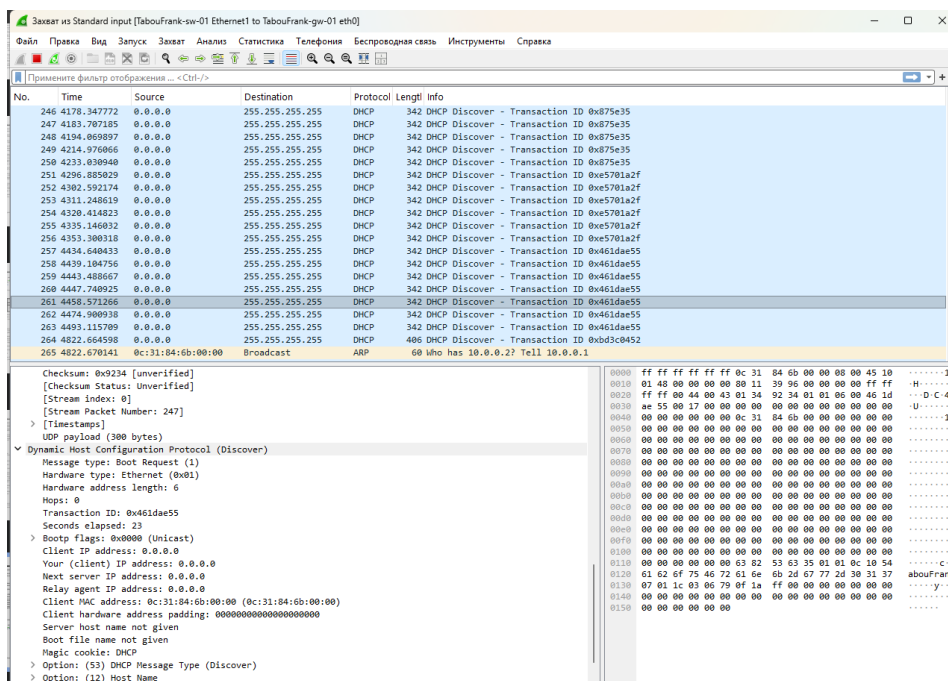


Рис. 13: Проверка IPv6-настроек на PC2

```

TabouFrank@TabouFrank-gw-01:~$ show log | grep dhcp
Jan 03 22:41:51 dhclient-script-vyos[1846]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:41:52 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:41:52 dhclient-script-vyos[1846]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:41:52 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:43:47 dhclient-script-vyos[2089]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:43:47 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:43:47 dhclient-script-vyos[2089]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:43:47 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:45:55 dhclient-script-vyos[2139]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:45:55 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:45:55 dhclient-script-vyos[2139]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:45:55 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:47:42 dhclient-script-vyos[2173]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:47:42 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:47:42 dhclient-script-vyos[2173]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:47:42 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:49:20 dhclient-script-vyos[2221]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:49:20 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:49:20 dhclient-script-vyos[2221]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:49:20 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:50:59 dhclient-script-vyos[2271]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:50:59 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:50:59 dhclient-script-vyos[2271]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:50:59 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:53:23 dhclient-script-vyos[2320]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:53:23 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:53:23 dhclient-script-vyos[2320]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:53:23 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:55:47 dhclient-script-vyos[2371]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:55:47 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:55:47 dhclient-script-vyos[2371]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:55:47 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:58:05 dhclient-script-vyos[2427]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:58:05 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:58:05 dhclient-script-vyos[2427]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:58:05 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:00:20 dhclient-script-vyos[2489]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:00:20 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:00:20 dhclient-script-vyos[2489]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:00:20 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:00:57 sudo[2607]: TabouFrank : TTY=ttys0 ; PWD=/home/TabouFrank ; USER=root ; COMMAND=/usr/bin/sh -c /usr
r/sbin/vysshm /usr/libexec/vyos/conf_mode/dhcp_server.py
Jan 03 23:00:57 vyos-confd[610]: Received message: {"type": "node", "data": "/usr/libexec/vyos/conf_mode/dhcp_se
rver.py"}
Jan 03 23:02:07 dhclient-script-vyos[2646]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:02:07 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:02:07 dhclient-script-vyos[2646]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:02:07 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:03:53 dhclient-script-vyos[2680]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:03:53 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:03:53 dhclient-script-vyos[2680]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:03:53 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:06:23 dhclient-script-vyos[2714]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:06:23 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:06:23 dhclient-script-vyos[2714]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client

```

Рис. 14: Ping IPv6-адреса маршрутизатора

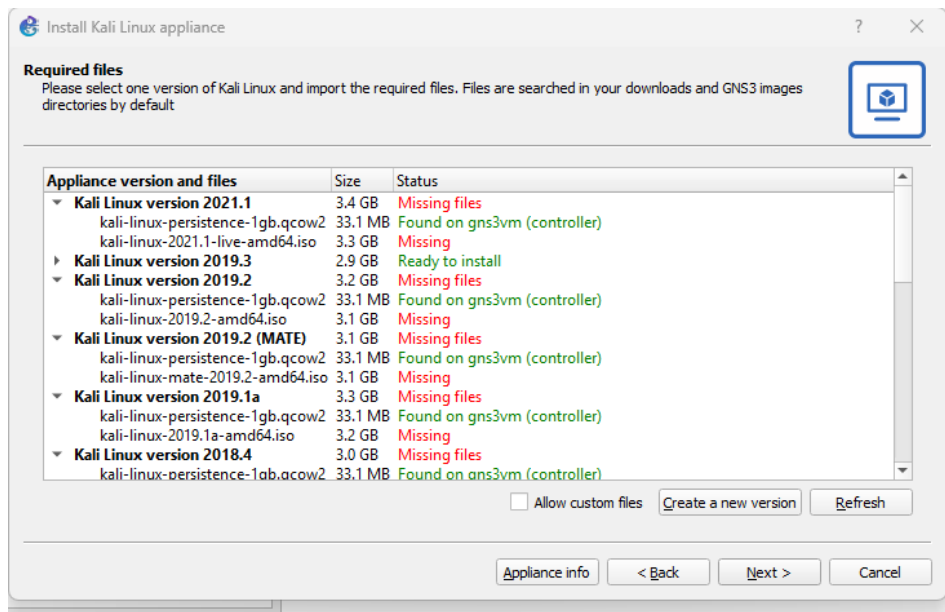


Рис. 15: Получение параметров по DHCPv6 Stateless

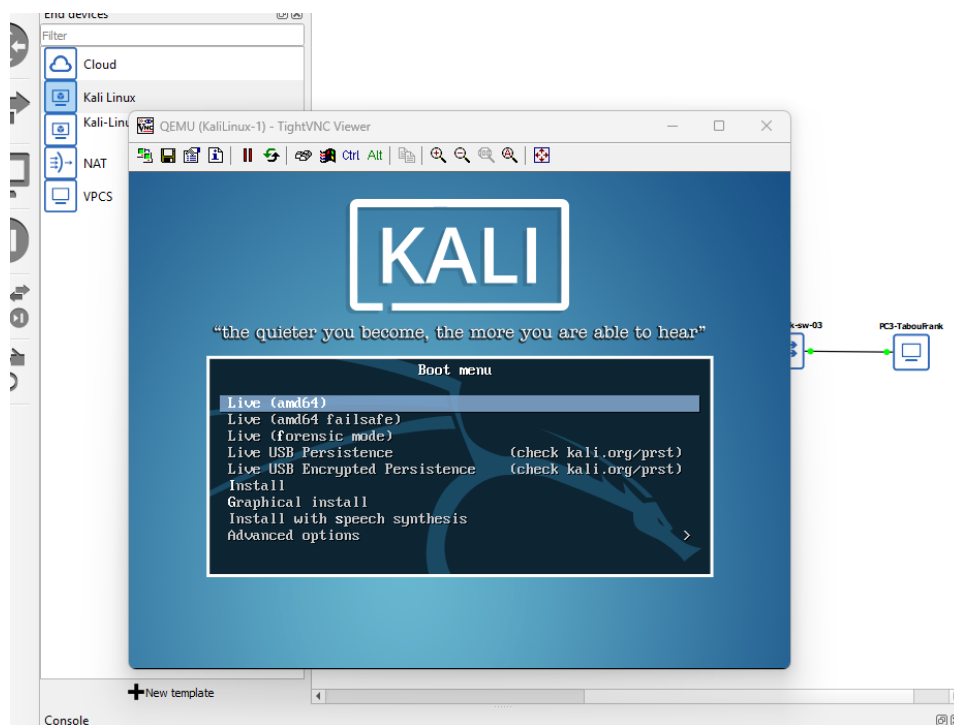


Рис. 16: Состояние аренд DHCPv6



Рис. 17: Анализ DHCPv6-пакетов в анализаторе трафика

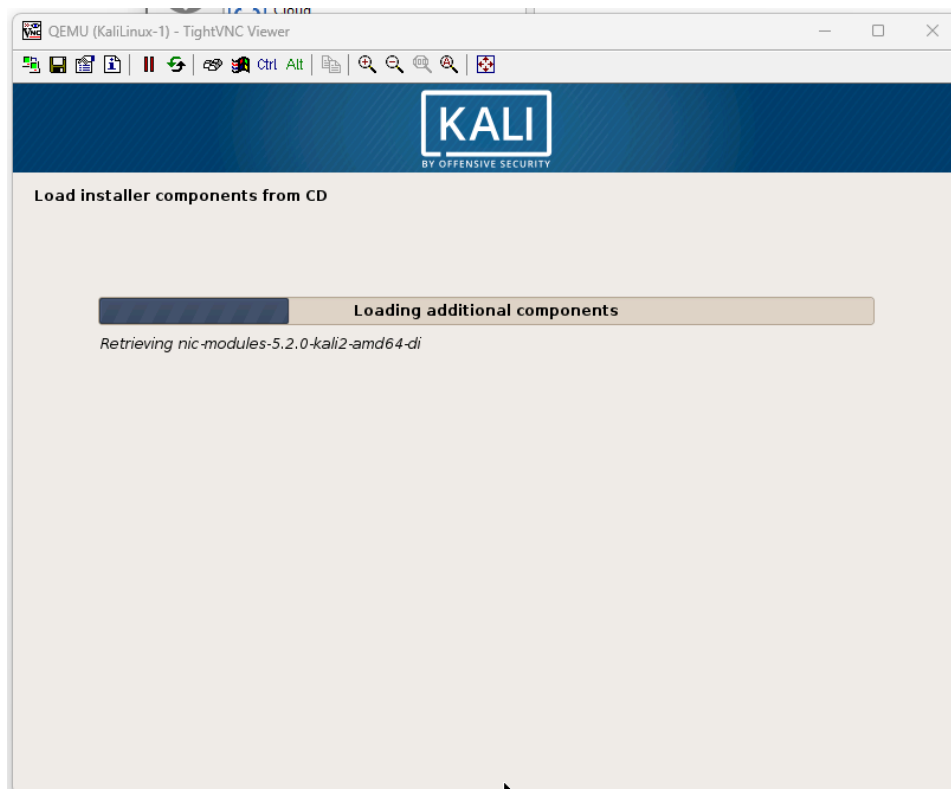


Рис. 18: Настройка DHCPv6 Stateful на VyOS

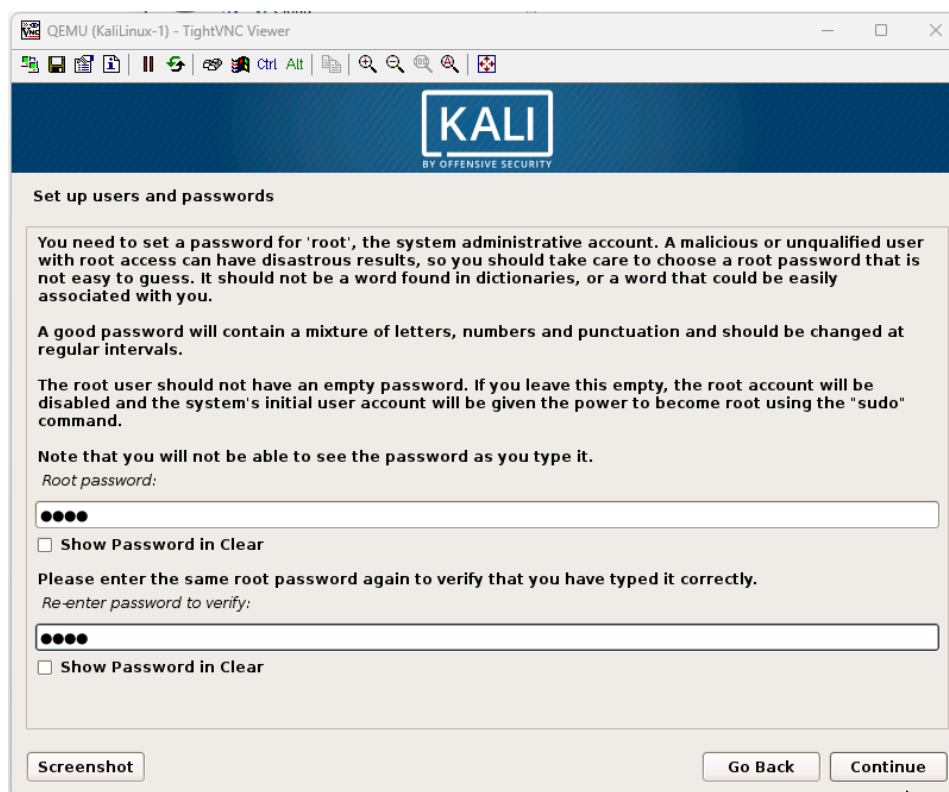


Рис. 19: Начальные IPv6-настройки на РСЗ

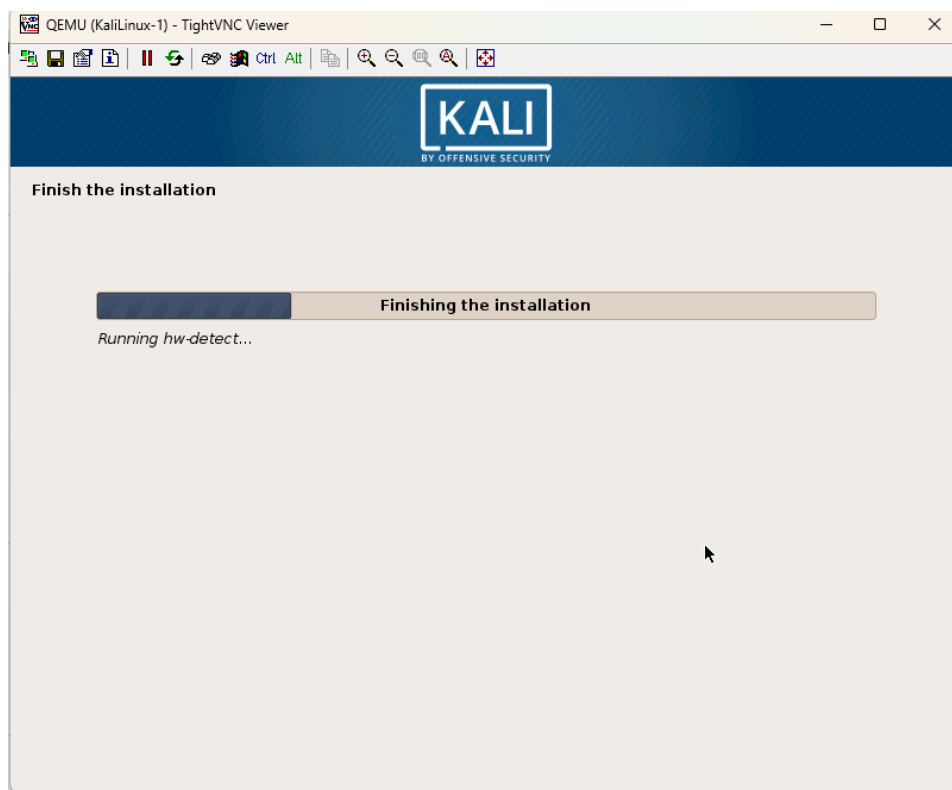


Рис. 20: Получение IPv6-адреса по DHCPv6 Stateful

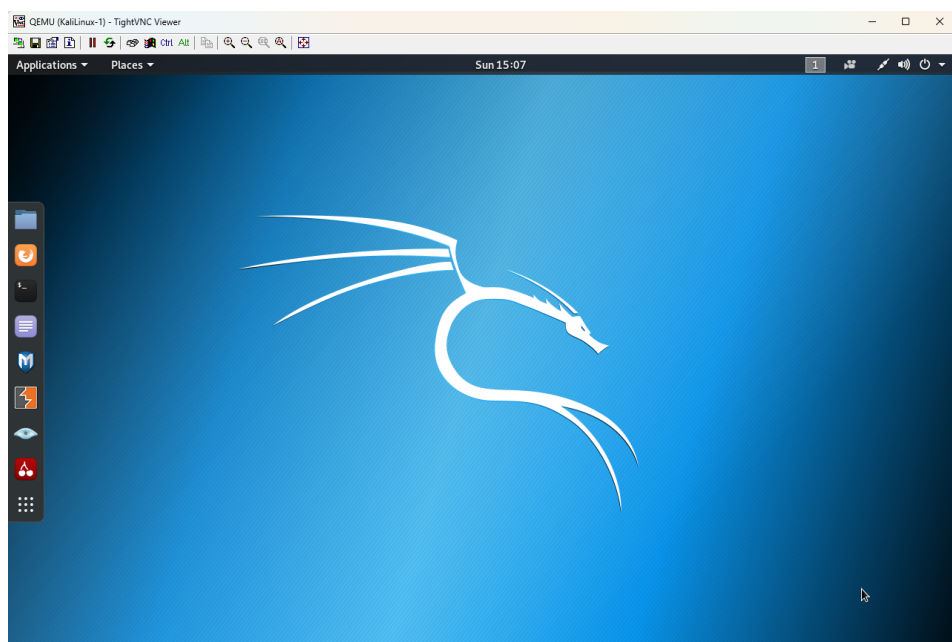


Рис. 21: Список аренд DHCPv6

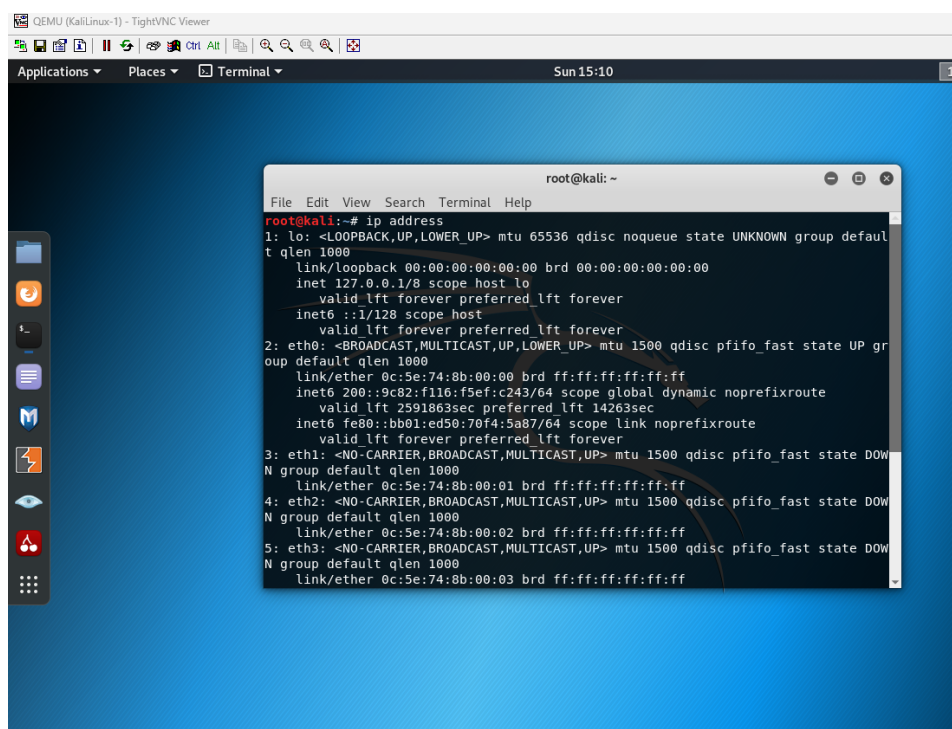


Рис. 22: Анализ DHCPv6-пакетов в анализаторе трафика